TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

Preparatory Processes on Cellulose, Protein and Man-made Fibres

The theory of TE-2205 put to work: desizing, scouring and bleaching cotton, scouring wool, degumming silk and scouring polyester — and why each fibre sets limits the others do not.

TE-2206 · TEXTILE CHEMISTRY & COLORATION-I SESSIONAL · EXPERIMENT 2 · LEVEL 2 TERM 2

1. Aim and theory

Aim: to carry out, on a laboratory scale, the preparatory sequence on three fibre families — desizing, scouring and bleaching of cotton (cellulose); gentle scouring of wool and degumming of silk (protein); scouring of polyester (man-made) — and to record the weight loss and absorbency change each stage produces. Grey material cannot be dyed level or bright: preparation removes everything that is not fibre, and this experiment is where that sentence becomes a set of baths.

What each bath must remove

FIBRE	IMPURITIES TO REMOVE	PROCESSES IN THIS EXPERIMENT
Cotton (cellulose)	Warp size (starch — an added impurity, up to ~10% on woven goods); natural waxes, pectins, proteins, seed-coat fragments, natural colour	Enzymatic desizing → alkaline scouring → peroxide bleaching
Wool (protein)	Wool grease (lanolin), suint (dried sweat salts), dirt and vegetable matter	Mild detergent scouring only — low temperature, near-neutral pH
Silk (protein)	Sericin (silk gum, ~20–25% of the raw silk) coating the fibroin	Degumming in near-boiling soap solution
Polyester (man-made)	Spin finishes, coning oils, knitting lubricants, dust; cyclic oligomer at the surface	Mild alkaline detergent scouring; bleaching normally unnecessary

The chemistry underneath

Desizing. Starch size is a film of amylose/amylopectin — water-insoluble as applied, so it survives weaving but blocks every later bath. Bacterial **α-amylase** cuts the starch chains at random into soluble dextrins and sugars that rinse away. The enzyme is a protein: it works in a window (about pH 5.5–7, 55–70 °C for the common laboratory grades) and is destroyed outside it.

Scouring. Hot dilute NaOH attacks cotton's natural impurities together: waxes are saponified, pectins are hydrolysed to soluble pectates, proteins are broken down — in short, saponifiable and alkali-soluble matter leaves the fibre:

$$(C_{17}H_{35}COO)_3C_3H_5 + 3\ NaOH \rightarrow C_3H_5(OH)_3 + 3\ C_{17}H_{35}COONa \quad \text{(wax ester} \rightarrow \text{glycerol + soap)}$$

The soap formed in the bath helps emulsify the wax that is not saponified; added detergent and wetting agent complete the job. What remains is nearly pure cellulose — hydrophilic, absorbent, and still slightly coloured.

Bleaching. Hydrogen peroxide bleaches through the perhydroxyl ion, released in alkali:

$$H_2O_2 + OH^- \rightleftharpoons HOO^- + H_2O$$

HOO^- destroys the conjugated double-bond systems of the natural colouring matter. Two controls decide success: the bath is held at pH 10.5–11 (enough alkali to form HOO^-, not so much that the peroxide wastes itself), and a **stabiliser** (sodium silicate or an organic stabiliser) suppresses the violent catalytic decomposition that iron, copper or rust spots would otherwise trigger — catalytic decomposition attacks the cellulose itself and is the classic cause of pin-holes and tendering in bleached cotton.

Protein fibres set their own limits. Keratin (wool) and fibroin (silk) are polyamides: strong alkali hydrolyses their peptide chains and, in wool, breaks the disulphide crosslinks that give the fibre its strength — wool is visibly damaged above pH ~10 and dissolves in hot strong alkali. Wool also **felts**: its surface scales ratchet the fibres together under heat, moisture, alkali and agitation. Chlorine bleaches are forbidden on both protein fibres — hypochlorite chlorinates and yellows wool and can dissolve it outright; silk is degummed, not bleached, at this stage, and peroxide for either fibre is used dilute, mild and cool.

Polyester is hydrophobic and almost pure as spun; preparation is a wash, not a chemical conversion. A mild soda-ash/detergent bath at 70–80 °C strips oils and finishes. Bleaching is skipped: the fibre is already white, and brightness is set later with an optical brightening agent during dyeing/finishing. Its caution is different — prolonged strong alkali hydrolyses the surface (the basis of deliberate “alkali deweighting”), so the scour stays mild.

2. Apparatus and procedure

- Thermostatic water bath (or hot plate with large beakers), beakers of 500–1000 mL, thermometer, glass rods, balance (0.01 g), measuring cylinders, sample pieces of grey cotton fabric, raw wool, silk fabric/yarn and polyester fabric (about 5–6 g each, weighed accurately).
- Chemicals: α-amylase, NaOH, soda ash (Na_2CO_3), non-ionic detergent, wetting agent, H_2O_2 (35%), sodium silicate, acetic acid, soap flakes.

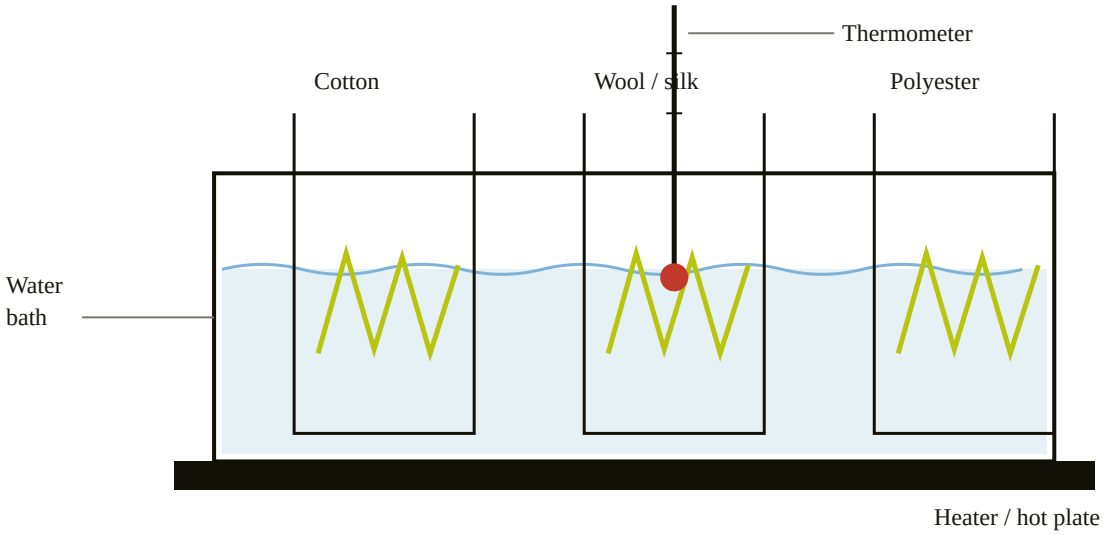


Fig. 1 — Laboratory bath set-up: one beaker per fibre, each at its own recipe, MLR 1 : 20, temperature watched continuously.

Recipes (laboratory scale, MLR 1 : 20 throughout)

BATH	RECIPE	PH	CONDITIONS
Cotton — desize	α-amylase 3 g/L, wetting agent 1 g/L	5.5–7	60–70 °C, 45 min
Cotton — scour	NaOH 3 g/L, detergent 1 g/L	~13	95–100 °C, 60 min
Cotton — bleach	H_2O_2 (35%) 4 mL/L, sodium silicate 3 g/L, NaOH to pH	10.5–11	90 °C, 60 min
Wool — scour	Non-ionic detergent 1.5 g/L, soda ash 0.5 g/L	≤ 9.5	45–50 °C, 30 min, minimal agitation
Silk — degum	Soap 5 g/L (≈ 25% on weight of silk)	9.5–10	90–95 °C, 60 min

Polyester — scour	Detergent 1.5 g/L, soda ash 1.5 g/L	~10–11	70–80 °C, 30 min

Procedure

1. Condition and weigh each sample accurately (W_0). Mark identity with a loose coloured sewing thread — pencil and ink do not survive the baths.
2. Run the cotton sequence stage by stage: desize → rinse → scour → rinse → bleach. After the bleach, rinse and **sour** (acetic acid 1 mL/L, 10 min, room temperature) to kill residual alkali, then rinse again. Between stages, squeeze the sample uniformly and weigh if stage-wise losses are wanted.
3. Run the wool and polyester scours and the silk degumming in parallel baths at their own temperatures; handle the wool with the least possible movement.
4. Dry all samples flat at room temperature (or below 60 °C), condition, and reweigh (W_1). Record weight loss at each stage and overall.
5. Test absorbency of the prepared cotton by the drop test (Experiment 3) and compare with the grey sample: the grey drop stands; the prepared drop should vanish almost at once.

FIBRE-SPECIFIC CAUTIONS — THE MARKS OF THIS EXPERIMENT

Wool: never strong alkali, never above ~50 °C in scouring, never chlorine bleach, never vigorous agitation — the penalties are dissolved keratin and irreversible felting.
Silk: degumming removes sericin by design (expect ~20–25% loss); exceeding time/temperature attacks the fibroin itself and the loss becomes damage.
Cotton: peroxide without stabiliser, or with rust contamination, tenders the cloth; desizing baths held too hot before the enzyme dissolves denature the amylase and the size stays.
Polyester: keep alkali mild and time short unless surface hydrolysis (deweighting) is actually intended.

3. Worked calculation (model data)

WEIGHT LOSS THROUGH THE COTTON SEQUENCE

$$\text{Weight loss } \% = (W_0 - W_1)/W_0 \times 100$$

Grey cotton W_0 = 6.00 g. After desizing: 5.55 g → loss = (6.00 − 5.55)/6.00 × 100 = **7.5%** (the size coming off).
After scouring: 5.28 g → stage loss = (5.55 − 5.28)/5.55 × 100 = **4.9%** (waxes, pectins, proteins).
After bleaching: 5.22 g → stage loss = (5.28 − 5.22)/5.28 × 100 = **1.1%** (colour bodies are a small

mass).
Overall loss = (6.00 − 5.22)/6.00 × 100 = 13.0% — and the absorbency changed from a standing drop (> 60 s) on the grey cloth to about 2 s on the prepared cloth. Big stage losses at desize and scour, a small one at bleach: that pattern is the signature of a healthy sequence. A large *bleach-stage* loss, by contrast, would signal cellulose attack, not better preparation.

The silk sample, for comparison, lost 22.4% in degumming (within the expected 20–25% sericin band) and the wool 6.8% in scouring (grease, suint and dirt) — losses that are the *purpose* of those baths, to be read against their expected bands, not against zero.

4. Precautions

- Keep MLR, temperature and time identical across repeats — a weight-loss comparison between baths of different liquor ratios proves nothing.
- Weigh only conditioned, uniformly squeezed samples; a wetter sample fakes a smaller loss.
- Never let the wool bath share alkali or agitation habits from the cotton line: separate glassware for the strong-alkali baths prevents expensive accidents.
- Check the bleach bath pH before immersing the cotton, and keep iron or rusty clips out of it — one rusty contact can catalyse a local peroxide burn.
- Rinse until the rinse water is neutral after scouring; residual alkali yellows cotton on drying and falsifies the bleach that follows.
- Amylase baths are made fresh and held inside the enzyme's temperature window; a boiled desize bath is just warm water with denatured protein in it.

5. Practice (viva preparation)

Answer in your own words; brief model answers follow.

1. Why is woven cotton desized before scouring, and why is the desize enzymatic rather than a hot alkali boil?
2. Write the saponification that occurs in cotton scouring, and name two other impurity classes the NaOH removes and how.
3. Why does the peroxide bath need both a controlled pH and a stabiliser? What fault does each prevent?
4. Explain, from the fibre's chemistry, three separate reasons the wool bath is mild, cool and quiet.
5. A polyester sample lost 11% in a “scouring” bath of 10 g/L NaOH at the boil. Is this good preparation? Explain.

6. A cotton sample weighs 5.60 g grey and 5.15 g after desizing and scouring. Calculate the combined loss and comment.

Answer key

1. The starch film is hydrophobic and sealed: scouring liquor cannot reach the fibre evenly through it, so scouring first would be patchy. Enzymatic desizing is selective — α-amylase hydrolyses starch to soluble sugars under mild conditions that leave the cellulose untouched; a long hot alkali boil would also attack it, less completely and at higher chemical and energy cost.
2. $(C_{17}H_{35}COO)_3C_3H_5$ + 3 NaOH → glycerol + 3 sodium stearate (soap). Also: pectins are hydrolysed to soluble sodium pectates, and proteins are hydrolysed to soluble amino-acid salts; unsaponified wax is emulsified by the soap and detergent formed/added.
3. pH 10.5–11 maximises the active HOO^- ion without wasting peroxide; too alkaline and the peroxide decomposes uselessly, too acid and it barely bleaches. The stabiliser suppresses metal-catalysed (Fe/Cu) radical decomposition, which otherwise attacks cellulose — causing strength loss and pin-holes.
4. (i) Keratin is a polyamide: alkali hydrolyses peptide bonds and breaks disulphide crosslinks, weakening and eventually dissolving the fibre — hence mild pH. (ii) Heat accelerates that hydrolysis and the felting reaction — hence ≤ 50 °C. (iii) Wool's surface scales ratchet together under agitation in liquor (felting shrinkage) — hence minimal movement.
5. No. Polyester carries only a percent or so of oils and finish; an 11% loss means the strong hot alkali hydrolysed the fibre surface itself (uncontrolled deweighting) — the sample is damaged, thinner and weaker, not “extra clean.”
6. Loss = (5.60 − 5.15)/5.60 × 100 = 8.0%. Consistent with size (~5–7%) plus part of the natural impurities — a plausible combined desize + scour figure for woven cotton.
